

SUBJECT: APPLICATIONS DEVELOPMENT THEORY (APT470S) Full time and part time	EXAM DATE: JUNE 2022	DURATION: 105 Minutes 90 min Writing time 15 min Reading and Upload time	TOTAL MARKS: 60
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Cape Peninsula
University of Technology
 creating futures

DEPARTMENT OF INFORMATION TECHNOLOGY
FACULTY OF INFORMATICS AND DESIGN

COURSE(S): DIPICTA: DIPLOMA IN INFORMATION AND COMMUNICATION TECHNOLOGY:
 APPLICATIONS DEVELOPMENT

EXAMINER: MS G KHAN
MODERATOR: MS F ALLIE
EXTERNAL MODERATOR: DR M NORMAN

GENERAL INSTRUCTIONS

- This is an individual test which should solely be done by you.
- You are not allowed to contact any person using any device or technique that **may compromise your individual work** when taking this assessment.
- You are required to answer all questions.
- The questions are presented one at a time without backtracking.
- All test questions relate to the scenario supplied below. Read it carefully before answering the questions.
- This test **counts 30%** of your final mark for this subject and covers work from terms 1 and 2 as per the topic list in the Subject Guide.
- This test is an open book assessment with pre-conditions. Carefully read and comply with the instructions below which differ from a physical assessment on campus.

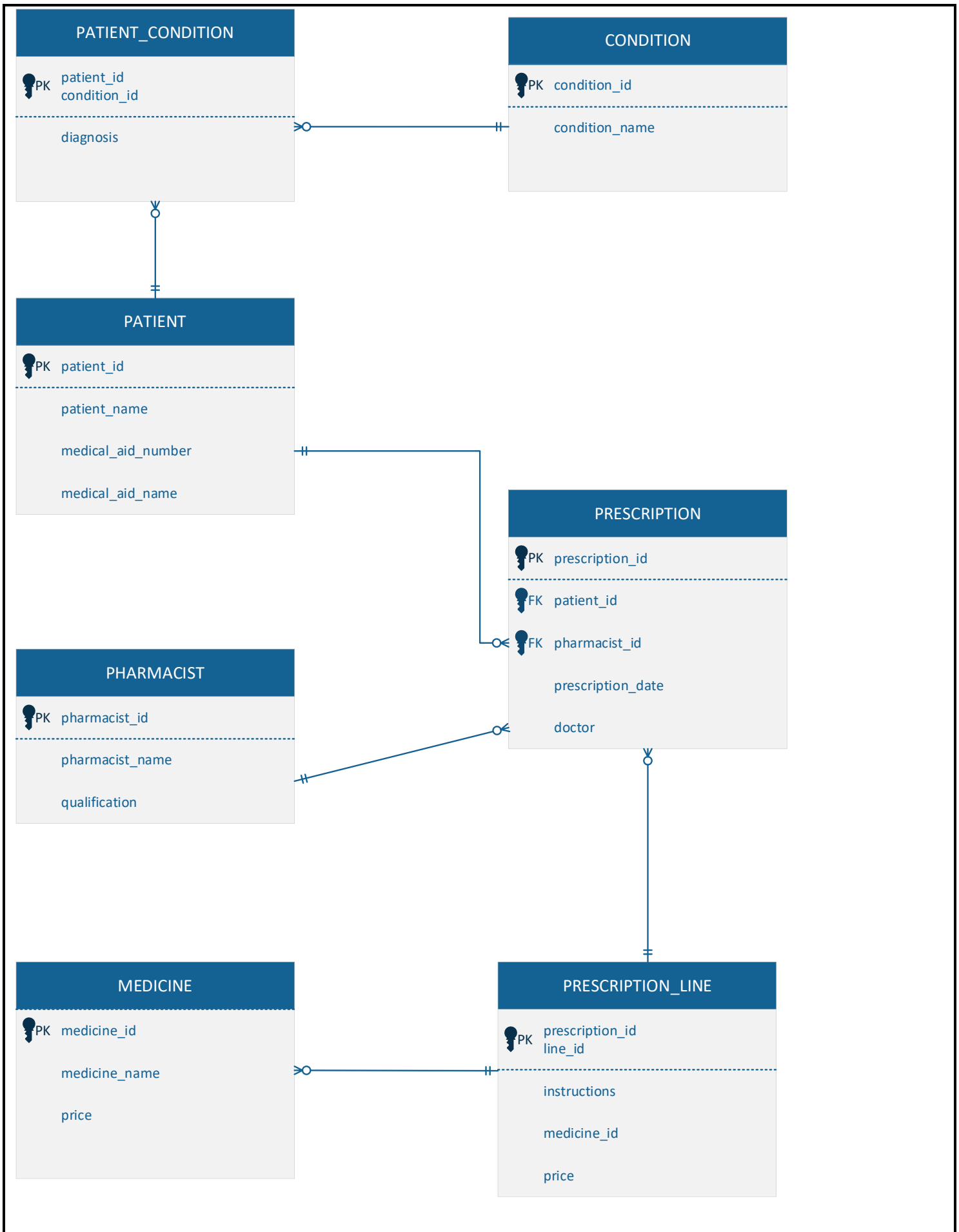
SPECIAL INSTRUCTIONS

- This **timed assessment** is 105 minutes in length . One and a half hours have been allocated for writing the test and 15 minutes have been allocated as reading and upload time. You are required to monitor the time during the assessment as the test will automatically terminate once the time has expired.
- This assessment requires a working device, a reliable internet connection and sufficient data/bandwidth for the duration of the assessment.
- Students who experience technical difficulties during the test session are required to email me a screen shot of their issue at khang@cput.ac.za immediately so that appropriate measures can be taken to address the problem.
- Students are reminded to do their own work. CPUT's assessment rules and regulations will apply to any form of plagiarism.
- **Do not** copy and paste answers from any resource. All answers should be expressed in your own words.
- Any resemblance of work copied from another source will be **flagged to the examiner, ignored and result in a mark of zero.**
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Scenario for the *GoodHealth Pharmacy*

Carefully examine the database design below for a system used by the **GoodHealth Pharmacy**. The system keeps track of conditions that patients have, as well as the medicines prescribed for each condition. The date when a patient was first diagnosed with a particular condition is kept in the PATIENT_CONDITION table. A pharmacist is always responsible for handling a prescription to ensure that the medicine is dispensed correctly. A prescription is always for one patient but can contain more than one prescribed medicine. The PRESCRIPTION_LINE table details each medicine on the prescription and the price is added at the time the medicine is dispensed to the patient.

The pharmacy has been in operation for many years and due to its location, caters largely to senior citizens. The current owner of the pharmacy is considering a few improvements to the system and "going online". Based on this scenario answer the following questions -



QUESTION 1 (15 MARKS)

1.1	The pharmacy owner has requested that the new system be of “high quality” even though the budget is limited. However, a focus on quality demands resources and time. Since you have decided to develop this system using an agile approach, can a quality focus still be maintained?	5
1.2	As part of the effort to modernize the system you decide that the system should be hosted on the Cloud. Prepare a list of reasons which explains why this could be a good option.	5
1.3	Determining the correct requirements is neither easy nor straightforward. Provide a list of 5 questions that you would ask when validating the requirements for the new pharmacy system.	5

QUESTION 2 (15 MARKS)

2	The pharmacy intends to fully automate the process of filling and delivering prescription medicines. The entire process will revolve around a Web-based application that is accessible by physicians (or their assistants), pharmacists, and patients. You need to only consider a small part of the work process for the situation where a patient asks for a refill of their prescription . You wish to create a swimlane activity diagram to help you in understand the associated workflow. Provide a step-by-step account of the tasks (and roles) involved in the workflow for this activity.	15
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QUESTION 3 (15 MARKS)

3.1	The software engineer should be able to recognize the key characteristics that will influence the design of the interface from the swimlane activity diagram. Discuss the characteristics you have recognized from your design.	6
3.2	Assume that you are developing a MobileApp to access the online pharmacy and while it will carry out functions typical of most pharmacies the app should also provide information about the medicines so that warnings of potential drug interactions can be issued. Discuss any 3 special usability tests that may be conducted for this MobileApp beyond concerns related to its appearance. Provide an example for each test category you suggest.	6
3.3	Provide an example for each of the “golden rules” listed below which will make the MobileApp more reliable and comfortable for users:- • <i>Place the user in control.</i> • <i>Reduce the user’s memory load.</i> • <i>Make the interface consistent.</i>	3

QUESTION 4 (15 MARKS)

4	Explain why choosing to develop the MobileApp for several devices is a costly design decision. Is there a way to mitigate the risks of supporting the wrong platform?	15
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